

Response to Reviewers

The Energy Cost of Privacy in Federated Learning
cSIS 2026 — Track 1, Green & Intelligent Computing

We thank both reviewers. Two of the findings are substantial and we have accepted them in full: the privacy guarantee was mislabelled (R2.3), and our first contribution rested on an unreconciled discrepancy between two measurement passes (R2.1). Both are now fixed in a way that makes the error impossible to repeat, and the second has become a result in its own right.

Where every number now comes from. The paper no longer contains a figure typed by hand. All tables and all in-text values are printed by the analysis script from the same arrays, on the committed measurements:

```
analyze.py results_har/ --latex --realised-eps spent_har.json          # Tables I, II, III
--targets 0.4,0.5,0.7,0.85                                           # CIFAR-10 check
analyze.py results/ --latex --realised-eps spent_cifar.json          # floors, hardware,  $\sigma$ 
analyze.py results_har/ --paper                                       # Section IV-E, HAR
analyze.py results_ablation_har/ --ablation                          # Section IV-E, CIFAR-10
analyze.py results_ablation/ --ablation                               # stopping rounds
per_seed_peak.py results_har/ <eps>                                   # participation audit
check_realised_epsilon.py results_har/ --worst-case                  # Figures 1 and 2
analyze.py results_har/ --no-peak-markers
```

The paper is now **6 pages**, verified by compilation, with no $\text{\LaTeX}$ warnings and no unresolved references.

1 Reviewer 1

The paper runs one page over the six-page limit.

Fixed: 6 pages. The cuts are the ones R2.6 identified — Section V-A reduced to its interpretive content, V-B’s restatement of IV-E removed, the carbon figure dropped because Table III is the same six numbers, captions shortened, and the overlap between the methodology’s limitations and Section V-E’s threats merged into one place. Section V-D, the measurement hazard (numbered V-E in the submitted version), is untouched; its three controls now cross-reference the methodology instead of restating it.

Uses “Index Terms” rather than “Keywords,” and sets the title in non-bold font. Also, figure and table captions are too long.

All three fixed. The keywords heading is renamed, the title is bold, and every caption is now at most three lines, with the numbers moved into the text where they can be read against the surrounding argument.

Two numbers in the reviewer’s summary of our contribution have changed, and we note them so the summary can be re-read against the corrected paper: the moderate-target cost is $5.7 \times$ ($1710/301 = 5.68$, as R2.2 computed), and the carbon factor is $18.8 \times$, now presented as context rather than as a result about privacy, for the reason R2.4 gives.

2 Reviewer 2

2.1 Two baselines and two noise floors for the same measurement

The reviewer is right, and the discrepancy had two independent causes.

The baselines. Figure 2 and the opening of Section IV were taken from a later re-run of the $\varepsilon = \infty$ condition alone, which overwrote the original sweep’s files — identical filenames, identical run-order seed — and was never committed. Table I was computed from the original sweep. We can defend only the committed data, so the re-run is gone and the whole paper now comes from the original sweep: baseline **13.98 J/round** (14.0 in Table I), mean overhead +51%, range +43.4% to +62.2%, board-power floor **7.80%**. The 14.5 J / +45% / 8.2% / 9.46% set is withdrawn.

The floor comparison was the wrong instrument, and this is the more important half of the reviewer’s point. Two things were wrong with it:

- The 12.5% spread is a spread in *net energy per round*, while the 7.80% floor is a spread in *board power*. The two differ by the round duration, which also varies across seeds. Every floor in the paper is now measured on the quantity it bounds. On energy per round the floor is ± 2.58 J/round, **12.0%** — not 7.8% — so the 12.5% spread sits essentially on it.
- Even with matched units, a floor is the spread of *one* condition’s seeds, whereas the claim is about the spread of *five condition means*, which with three seeds each is narrower by about $\sqrt{3}$. So we test the claim instead of comparing it to a floor: permuting the seed labels across the five private budgets gives $F = 1.46$, $p = 0.27$ on HAR and $F = 0.46$, $p = 0.79$ on CIFAR-10.

On the specific contrast the reviewer raises, $+43.4\%$ at $\varepsilon = 1$ against $+62.2\%$ at $\varepsilon = 4$: it is not resolvable. The three seeds of the $\varepsilon = 4$ condition span 20.9 to 24.1 J/round on their own, and the overheads are not ordered in ε ($\varepsilon = 8$ sits below $\varepsilon = 4$).

Section IV-A is retitled *The Per-Round Cost Does Not Track the Budget* and states the weaker claim the data supports: no resolvable dependence on ε , and explicitly **no claim of equality** — three seeds per budget bound a budget effect at about the size of the overhead’s own scatter, not at zero.

2.2 Headline figures not reproducible from the tables

Every one is corrected, and the text values are now printed from the same arrays as the tables rather than transcribed.

Claim	Submitted	Corrected
ratio at the 0.50 target	1.17–2.10	1.28–1.79 (as the reviewer computed)
moderate-target cost	$5.8\times$	$5.7\times$ ($1710/301 = 5.68$)
non-private at 0.85	878 J, 57.7 rounds	839 J, 57.0 rounds
post-peak energy, HAR $\varepsilon = 0.5$	2477 J over 83 rounds	withdrawn (see 4)
ablation per-round at $\varepsilon = 1$	29 J	22 J

On the 2477 J: the reviewer’s arithmetic is what exposed it. 2477 J over 83 rounds is 29.8 J/round against Table I’s 20.3 J net, because that number came from the withdrawn pass. On the committed data HAR has no resolvable stopping round at any budget, so the claim is gone rather than rescaled.

On the ablation’s 29 J: the corrected value is 22.03 ± 0.36 J/round, and the residual 2 J against Table I’s 20.05 ± 0.39 is a genuine between-session offset — the ablation sweep ran separately, with its own idle baseline. It is larger than either sweep’s across-seed spread, so we now state in Section IV-E that ablation energies are compared *within* the ablation sweep only, and we make no cross-sweep energy comparison anywhere.

On gross versus net: stated once in Section III-C (“every energy figure in this paper is net of the idle baseline unless a column is labelled Gross”) and again at the head of Section IV. Table I labels its gross column; every other energy in the paper is net.

2.3 The guarantee is mislabelled, and q_c assumes what we disclaim

Accepted in full, on both halves.

The label. Opacus clips and noises per-sample gradients inside local training, so the guarantee is **example-level**, applied locally, with the budget held per client over the whole run. It is not the client-level (user-level) DP of [2] and [11], which noises the entire client update and conceals participation itself. We have relabelled the abstract, Section I, Section II-D, Section III-B, Section V-E and the conclusion, and Section III-B now states the relation explicitly: our guarantee is weaker, and the per-round overhead we measure is the per-sample gradient work that both mechanisms need, so the energy result is orthogonal to the choice between them.

The q_c factor. Also accepted: q_c makes s_{total} an expected participation count, nothing in the protocol bounds the realised one, and the reported ε is therefore **optimistic rather than an upper bound**. We had begun auditing this before the reviews and the audit agrees with the reviewer in every run. Recomputing each client’s spent ε from the participation logs, the worst client overspends in **all 29 private runs**, by 4% to 22%:

target ε	0.5	1	2	4	8
worst spent, UCI-HAR	0.570	1.122	2.285	4.250	9.112
worst spent, CIFAR-10	0.611	1.162	2.251	4.356	8.850

Table I now carries a *Worst ε* column with the HAR values, and Section III-B states the remedy the reviewer names: a guarantee that must hold unconditionally has to drop q_c and calibrate against $RE \lceil n/B \rceil$, or cap participation per client. The audit script has a `--worst-case` mode that prints the ε of participation in every round, which is the figure such a calibration would have to use.

We would rather report this than hide it: the same optimistic calibration is common in federated DP simulations, and the audit is a few lines of logging plus a replay.

2.4 Two results carry less information than their framing

Accepted on both.

The post-peak energy share. The reviewer is right that with per-round energy flat, “83% of the joules” is 83 of 100 rounds restated, and right that the declines defining those peaks sat below our own floors. A peak is now reported only where the peak-to-final drop clears that condition’s across-seed spread in final accuracy *and* the individual seeds agree on where the peak falls. Under that test:

- **HAR fails at every budget.** The largest drop in a mean curve is 0.032 at $\varepsilon = 1$ against a ± 0.042 spread, and the per-seed peak rounds disagree (10, 33, 39 at $\varepsilon = 1$; 16, 54, 65 at $\varepsilon = 0.5$). The HAR stopping-round claim is withdrawn and Figure 1 is redrawn with no marker, its caption saying why.
- **CIFAR-10 passes at the two tightest budgets.** $\varepsilon = 0.5$: drop 0.026 against ± 0.010 , seeds peaking at rounds 8, 8 and 9. $\varepsilon = 1$: drop 0.044 against ± 0.014 , seeds between 10 and 16. At $\varepsilon \geq 2$ the drops fall inside their spreads and we claim nothing.

Section IV-C now leads with the peak’s position — round 8 of 30 at $\varepsilon = 0.5$ — and says in the text that the energy fraction tracks the fraction of the schedule after the peak precisely because per-round energy is flat, so the joules price the tail of a round budget the designer chose. The monotone peak ordering 7/11/19/24 is withdrawn: only two budgets resolve a peak at all.

The carbon factor. Also accepted. The ratio is the two grid intensities and would be identical for non-private training, and the paper now says so in Section IV-F. The abstract and Section IV-F lead instead with the quantity that is about privacy: the premium attributable to privacy alone, 106–148 gCO₂eq per 1000 runs on Algeria’s grid against 5.7–7.9 g in Norway. The geographic point is framed as context for a federated deployment, which cannot relocate the way a data centre can.

2.5 Reference verification and IEEE formatting

Thank you for checking these. All are fixed.

- **One author plus “et al.” where six are permitted.** This was IEEEtran’s default name limit rather than per-entry data. An @IEEEtranBSTCTL entry now sets the limit to six names and the manuscript cites it before the first citation; eight entries now show six authors before “et al.”
- **Missing volume, issue and pages.** The entries that lacked them were preprints typed as journal articles, which is why the fields were empty. The seven preprints are now @misc entries carrying the arXiv identifier, which IEEEtran prints, so nothing claims a volume it does not have. Two IEEE entries without volume or pages are genuinely early access and are now marked as such.
- **Institutional authors** (NVIDIA, Ember) were being initialised to “N. Corporation” and “E. Ember”; both are now braced.
- **[16] Anguita et al.** carried a spurious “and others” — the paper has exactly five authors — and the venue was abbreviated to “Esann”. Both corrected, and the entry is cited by its ESANN paper number (es2013-84) rather than by the page range that circulates in secondary sources, which we could not confirm against the proceedings.
- **[24] Aroussi et al.**, the DOI you asked for: 10.1109/IRASET68627.2026.11538879. It is now in the entry and prints in the reference list. We agree it had to be solid, since it carries the closest competing claim in the paper.

2.6 Page limit

The submission is now 6 pages, and we took the cut where the reviewer directed it. Section V-A is reduced to its interpretive content, V-B no longer restates Section IV-E, the carbon figure is dropped because Table III carries the

same six numbers, the captions are shortened, and the duplication between the methodology’s limitations and Section V-E’s threats is merged. The bibliography is tightened in form rather than shortened, since every entry is cited.

Section V-D, the accountant-in-the-measurement-window hazard (V-E in the submitted version), is untouched, and we have made it cheaper to act on: its three controls now point to the methodology that implements them, so the finding and its remedy are each stated once.

3 Summary of changes

Location	Change
Abstract, §I, §II-D, §III-B, §V-E, Conclusion	guarantee relabelled example-level, applied locally, per-client budget over the run
Abstract, §IV-A, §V, Conclusion §IV opening §IV-A	CIFAR-10 per-round overhead $+60\% \rightarrow +34\%$; permutation-test p -values added floors restated on the quantity each bounds; CIFAR-10 power floor $6.38\% \rightarrow \mathbf{8.9\%}$ retitled; floor comparison replaced by a permutation test; the $\varepsilon = 4$ contrast addressed
§III-C, §IV opening	gross versus net stated explicitly
Table I	<i>Worst ε</i> column added; caption shortened
Table II	$\varepsilon = 0.5$ at the 0.50 target shown as 334 J (12.0) $\ddagger$, two of three seeds, instead of hidden
§IV-B	$1.17\text{--}2.10 \rightarrow \mathbf{1.28\text{--}1.79}$; $5.8\times \rightarrow \mathbf{5.7\times}$; $878\text{ J}/57.7 \rightarrow \mathbf{839\text{ J}/57.0}$
§IV-C	rewritten: HAR stopping round withdrawn, CIFAR-10 resolvable at $\varepsilon \leq 1$, peak ordering withdrawn, energy share reframed
§IV-D	$1482\text{ J} / 406\text{ J} / 3.65\times / 0.503 \rightarrow \mathbf{1076\text{ J} / 677\text{ J} / 1.59\times / 0.490}$
§IV-E	six ablation values corrected; between-sweep offset stated; accuracy floors used for accuracy claims
§IV-F, Table III	premium leads; ratio $21.4\times \rightarrow \mathbf{18.8\times}$, framed as the grids’ own; intensities stated in the caption
§II	“83% of a run’s joules” replaced by the peak’s position
Figures	carbon figure dropped; Fig. 1 redrawn without a stopping marker; captions shortened
Bibliography	six-name limit, preprints as @misc with arXiv ids, braced institutional authors, Anguita corrected, Aroussi DOI added